

7/10

→ non isothermal film

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Quiz #3

$$\left(\frac{T}{T_1}\right) = \left(\frac{r}{r_1}\right)^n$$

$$\frac{D_{AB}}{D_{AB,1}} = \left(\frac{T}{T_1}\right)^{3/2} \therefore \frac{D_{AB}}{D_{AB,1}} = \left(\frac{r}{r_1}\right)^{3n/2}$$

Design Eqn: $\frac{1}{r^2} \frac{d}{dr} \left(r^2 D_{AB} \frac{dC_A}{dr} \right) = 0$

→ 1 dimensional, steady state, Spherical.

$$r^2 D_{AB} \frac{dC_A}{dr} = A$$

$$r^2 \left(D_{AB,1} \left(\frac{r}{r_1}\right)^{3n/2} \right) \frac{dC_A}{dr} = A$$

$$D_{AB,1} \frac{r^2 \cdot r^{3n/2}}{r_1^{3n/2}} \frac{dC_A}{dr} = A$$

$$D_{AB,1} \frac{r^{2+3n/2}}{r_1^{3n/2}} \frac{dC_A}{dr} = A$$

$$\frac{dC_A}{dr} = \frac{A r_1^{3n/2}}{D_{AB,1}} \left(\frac{1}{r^{2+3n/2}} \right)$$

$$2 + \frac{3n}{2} = \frac{4+3n}{2}$$

$$C_A(r) = \frac{A r_1^{3n/2}}{D_{AB,1}} \left(\int \frac{1}{r^{2+3n/2}} dr \right) + B$$

$$u = r^{\frac{4+3n}{2}}$$

$$r^{\left(\frac{4+3n}{2}\right)} = -\frac{4+3n}{2} r^{\frac{2}{2}}$$

$$C_A(r) = \frac{A r_1^{3n/2}}{D_{AB,1}} \frac{r^{-\left(\frac{2+3n}{2}\right)}}{-\left(\frac{2+3n}{2}\right)} + B$$

$$\downarrow$$

$$r^{-\left(\frac{2+3n}{2}\right)} = -\frac{2+3n}{2}$$

$$\frac{r}{-\left(\frac{2+3n}{2}\right)}$$

$$C_A(r) = \frac{-A r_1^{3n/2}}{D_{AB,1}} \frac{r^{-\left(1 + \frac{3n}{2}\right)}}{\left(1 + \frac{3n}{2}\right)} + B$$

$$BC-1: C_A(r_1) = C_1$$

$$BC-2: C_A(r_2) = C_2$$

$$C_1 = - \frac{A r_1^{3n/2}}{D_{AB,1} \left(1 + \frac{3n}{2}\right)} r_1^{-(1 + \frac{3n}{2})} + B$$

$$C_2 = - \frac{A r_2^{3n/2}}{D_{AB,1} \left(1 + \frac{3n}{2}\right)} r_2^{-(1 + \frac{3n}{2})} + B$$

Solve for A & B.

Expression for x_A ?
 $N_A = ?$